

INDIVIDUAL PROGRESS REPORT

Task 9: Integrated Pipeline, Algorithm Metrics, and Professional Visualizations

Group Topic	Boardie Analytics: A Big Data Approach to Student Boarding House Discovery and Geofenced Presence Pattern Analysis
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Lab No.	Task 9
Git Repo / Colab Link	https://github.com/carlpoot/week-4-biggs-data

Objectives

The objective of Task 9 is to extend the previous Boardie Analytics algorithm work into a more complete analytics pipeline that demonstrates integration with external tools and produces professional visualizations. The task focuses on preparing dashboard-ready processed data, exporting algorithm metrics, creating visual summaries, and organizing the outputs for GitHub submission and reporting.

Tasks Completed

- Reviewed the Task 9 requirements and connected the new activity to the previous Week 3 Random Forest classification model.
- Prepared an integrated analytics pipeline that loads the processed Boardie dataset and creates derived fields for dashboard use, including availability label, review quality, amenity level, and listing size.
- Continued the classification approach by training a Random Forest model to classify the listing price band using property, room, availability, review, and listing characteristic features.
- Fine-tuned the model through GridSearchCV using number of trees, maximum depth, and minimum samples per leaf as the main tuning parameters.
- Generated algorithm metrics such as accuracy, weighted precision, weighted recall, weighted F1 score, a classification report, and a confusion matrix.
- Exported processed files for external visualization tools, especially a dashboard-ready CSV that can be imported into Tableau, Power BI, Looker Studio, or spreadsheet-based dashboards.
- Created professional visualization outputs including price band distribution, room type by price band, availability by price band, feature importance, confusion matrix, neighbourhood median price, and association-rule lift charts.

- Added a simple association-pattern summary using support, confidence, and lift to align the output with the Chapter 9 discussion on association rule mining.
- Prepared a browser-based dashboard preview in HTML to demonstrate how the charts and metrics can be presented professionally.
- Saved the code notebook, Python script, processed CSV outputs, visualization files, README, and individual progress report for GitHub upload.

Tasks to be Completed

- Upload the Task 9 notebook, script, processed CSV files, visualization outputs, and report to the group GitHub repository.
- Replace the placeholder GitHub link in the progress report after the repository has been created and the files have been uploaded.
- If required by the instructor, import the dashboard-ready CSV into Tableau or another external tool and recreate the charts as an interactive dashboard.
- Continue improving the model by testing additional algorithms or adding more contextual features if the project dataset is expanded.

Pipeline Summary

Pipeline Stage	Output / Purpose
Processed Dataset	Uses the Week 3 processed Boardie analytics dataset as the base input.
Feature Preparation	Creates model-ready and dashboard-ready variables such as review quality, availability label, and amenity level.
Model Training	Trains a Random Forest Classifier to classify listing price bands.
Algorithm Metrics	Exports accuracy, weighted precision, weighted recall, weighted F1 score, classification report, and confusion matrix.
External Tool Integration	Exports tableau_boardie_dashboard_data.csv and other CSV summaries for Tableau, Power BI, Looker Studio, or spreadsheet dashboards.
Professional Visualizations	Creates PNG charts and a dashboard_preview.html file for presentation-ready viewing.

Algorithm Metrics Summary

Metric	Value
Accuracy	0.7200
Weighted Precision	0.7278

Weighted Recall	0.7200
Weighted F1 Score	0.7049
Training Rows	750
Testing Rows	250
Best Parameters	{"model__max_depth": 10, "model__min_samples_leaf": 1, "model__n_estimators": 100}

Results and Analysis

Task 9 delivered results for both model evaluation as well as professional visualization. In order to classify listings to specific price categories using machine learning, a Random Forest Classifier model was employed, with listing and availability features as inputs. After hyperparameter tuning, this model achieved an accuracy score of 0.7200 and a weighted F1 score of 0.7049, indicating that it provides a reasonable baseline to predict which listing belongs to what price category, even if the budget and premium price categories are less common compared to mid-range and upper-mid listings.

Finally, the CSV files that have been exported illustrate the integration aspect of the task. Rather than having the results displayed inside the notebook, these files can be opened in external tools such as Tableau, Power BI, Looker Studio, or spreadsheets.

These are the visualization diagrams for the results obtained after data analysis. The price range and room type visualizations depict the distribution of the listings, while the listing availability and the feature importance diagrams show the patterns that influenced the prediction made through the model. The confusion matrix indicates where the predictions were either right or wrong. The association pattern results are also related to the chapter 9 discussions, as they depict co-occurrence rules through support, confidence, and lift measures.

Figure 1. Listing distribution by price band

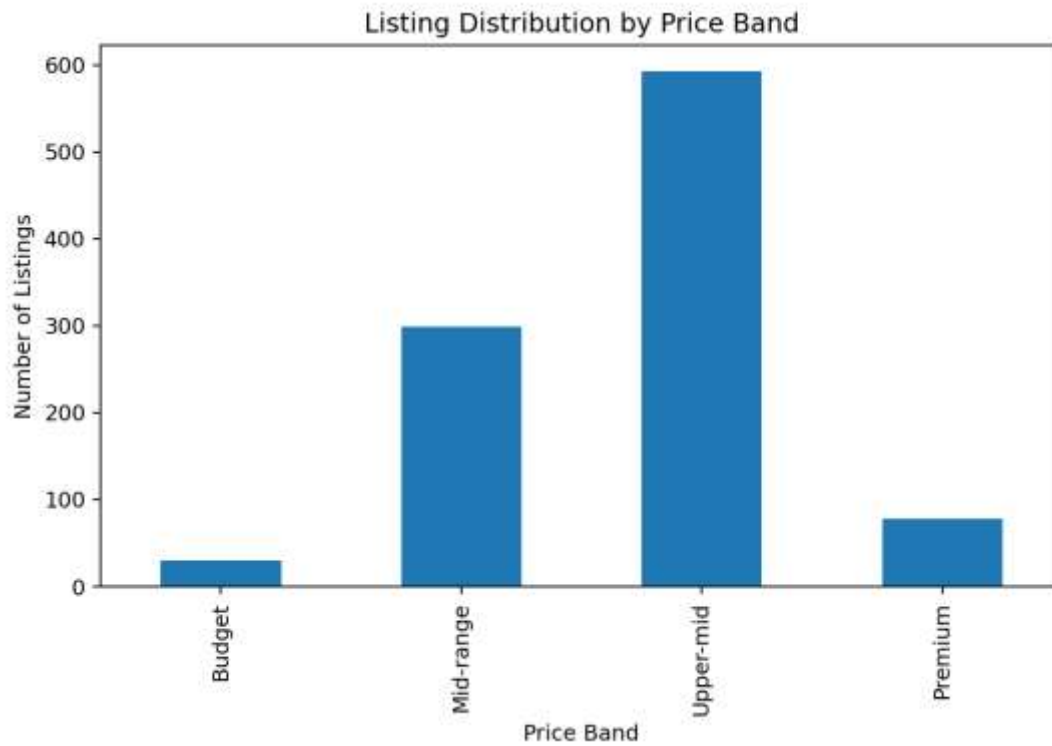


Figure 2. Price band composition by room type

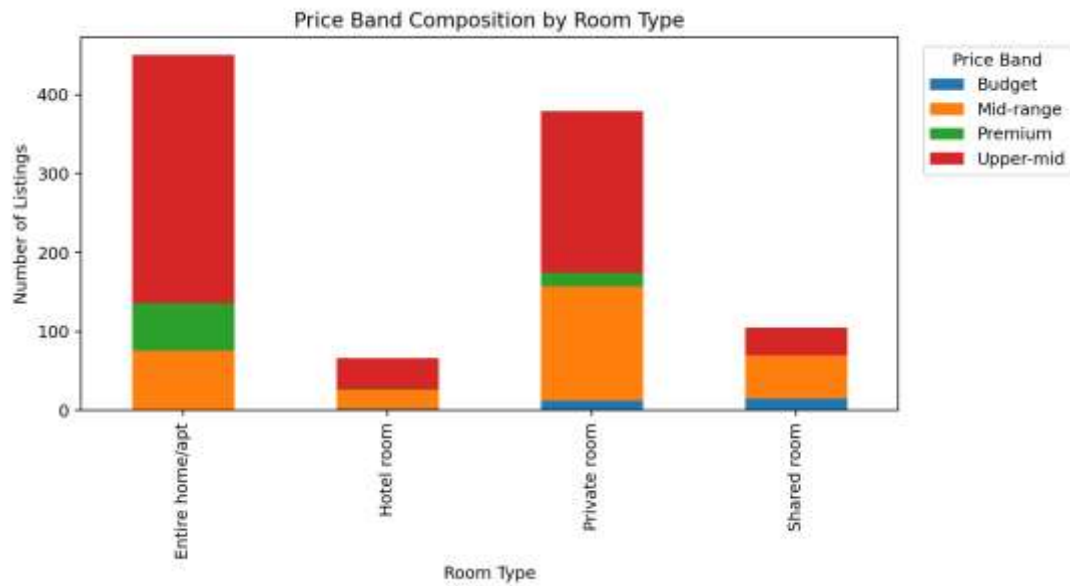


Figure 3. Average availability rate by price band

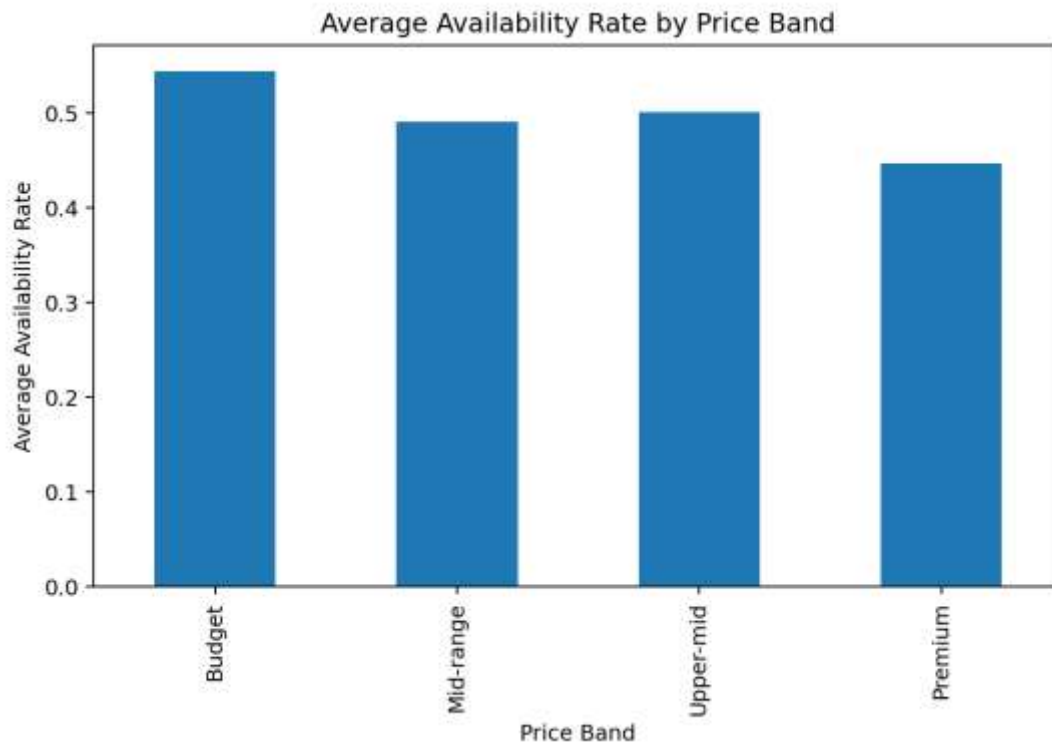


Figure 4. Feature importance for price-band classification

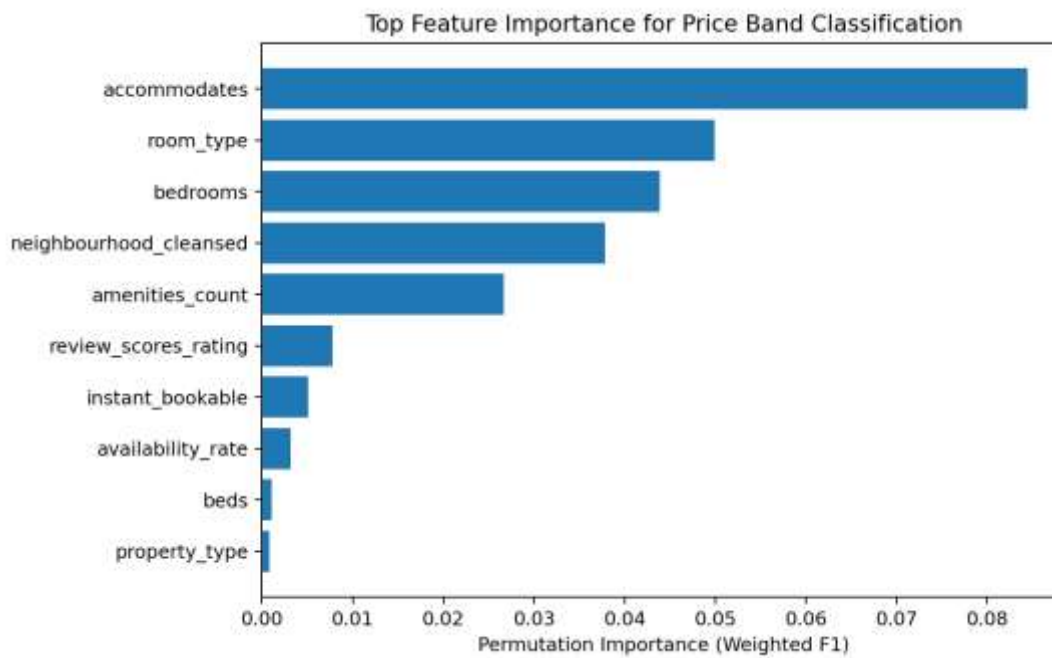


Figure 5. Random Forest confusion matrix

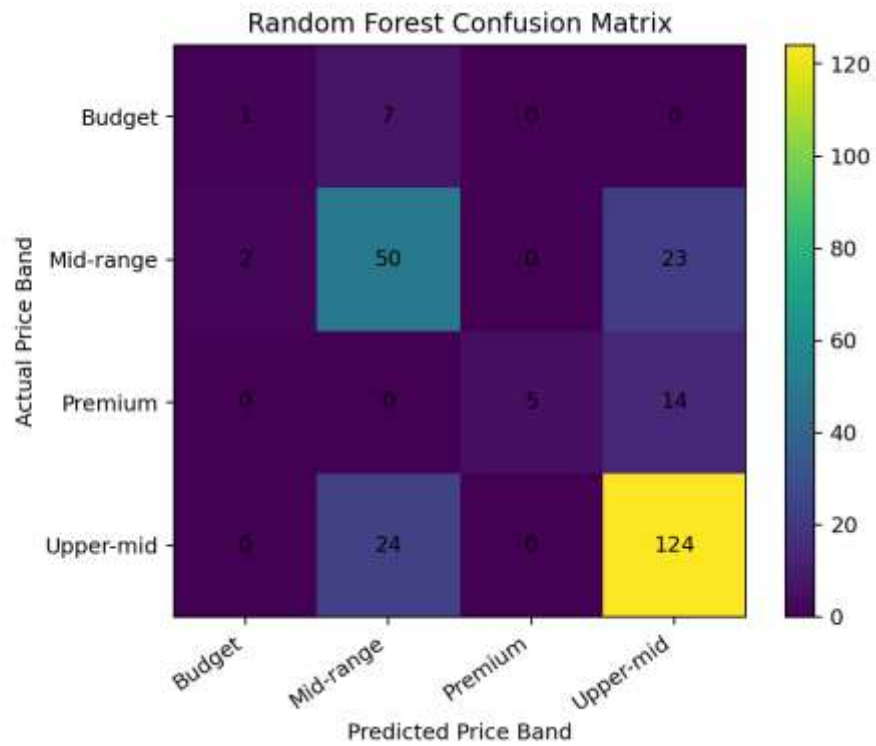


Figure 6. Top areas by median listing price



Challenges and Solutions

Challenge	Solution
The task requires external tool integration, but the group may not have a finalized Tableau workbook yet.	The pipeline exports dashboard-ready CSV files and an HTML dashboard preview so the output can be imported into Tableau or another BI tool later.
Some price-band classes are less frequent than others, which can affect classification performance.	The model uses stratified train-test splitting and class-weight balancing to reduce the effect of uneven class distribution.
Professional visualization requires more than basic exploratory charts.	The output includes cleaned charts with titles, labels, dashboard-ready summaries, and a browser preview for presentation use.
Chapter 9 discusses unsupervised association rule mining, while the previous task used classification.	A simple association-pattern summary was added using support, confidence, and lift while still preserving the previous Random Forest model metrics.

Conclusion

Objectives in Task 9 have been achieved through the extension of the previous Boardie Analytics Model to an analytics pipeline approach. This includes model performance metrics, optimized model classification output, dashboard-ready CSV export files, professional-level graphs and charts, HTML dashboard prototype, and project progress update. The exercise demonstrates how data that has undergone processing can be transitioned from coding analysis to external visualization tools.